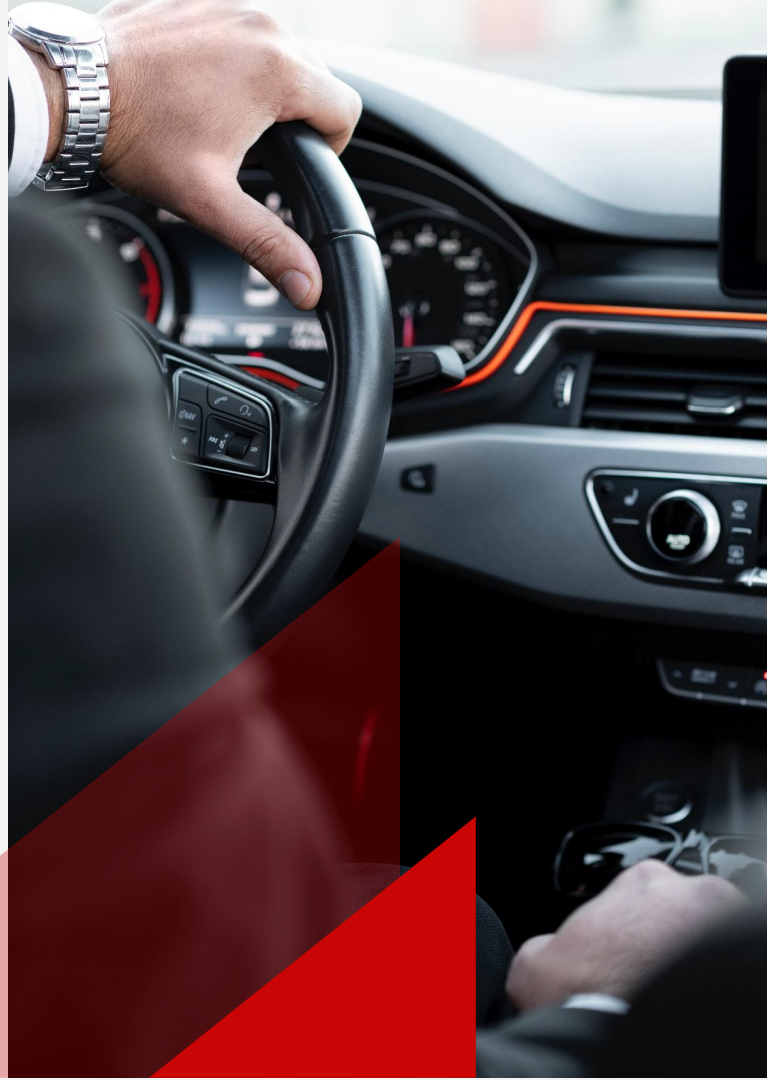
Fraud detection in car insurance claims

MISTI GTL - AI Uruguay 2026

Florencia Barrios, Maria Ines Martinote, Florencia Lucas



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Real-World Scenarios

1. Professional Fraud
Destroying luxury cars with hammers to collect high-value payouts.

2. Staged Accidents
Forcing rear-end crashes to blame innocent drivers.

3. Opportunistic Fraud
Buying a policy after an accident to claim pre-existing damage.

Fraud = Invisible Profit Loss

The Problem

For insurance companies manual fraud detection is slow, expensive, and prone to human error.

The Solution

A binary classification model trained on historical claim data.

How?

- **Input:** historical claims including policy details, vehicle category, and accident behavior.
- **Model:** CatBoost optimized with Optuna.
- **Output:** A binary prediction (fraud vs. legitimate) to prioritize high-risk investigations.

Data Overview

Source: Kaggle "Fraud Oracle" dataset

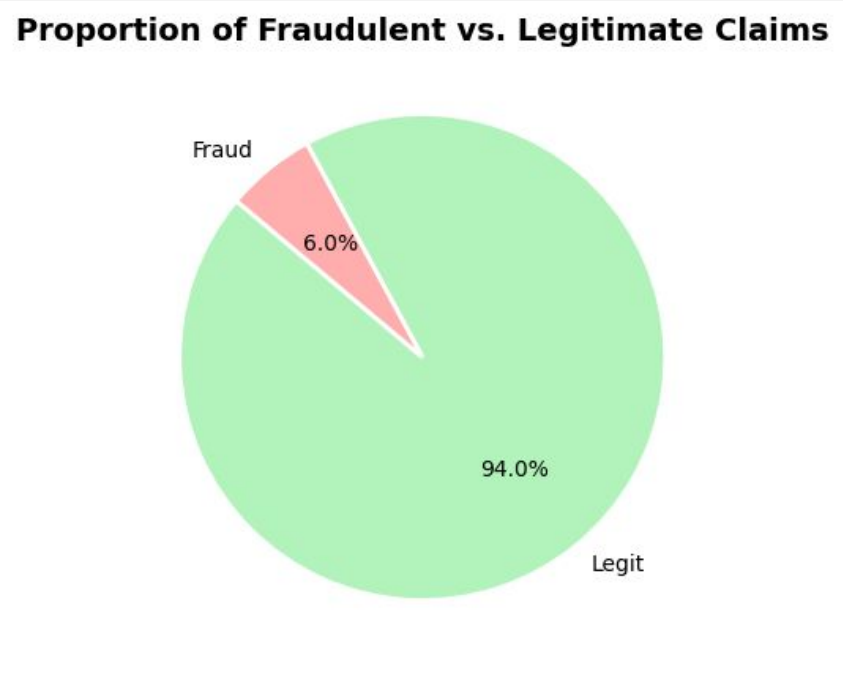
Volume: 15,420 rows and 33 features

Target: FraudFound_P (Binary: 0 or 1)

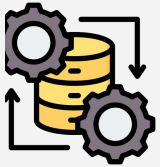
Limitation: Historical data (1994-1996) with no specific claim timestamps

Key Features:

- Policy (BasePolicy, Deductible)
- Vehicle & Driver: VehiclePrice, VehicleCategory, AgeOfPolicyHolder, DriverRating.
- Incident Details: Fault, AccidentArea, PoliceReportFiled.



Preprocessing



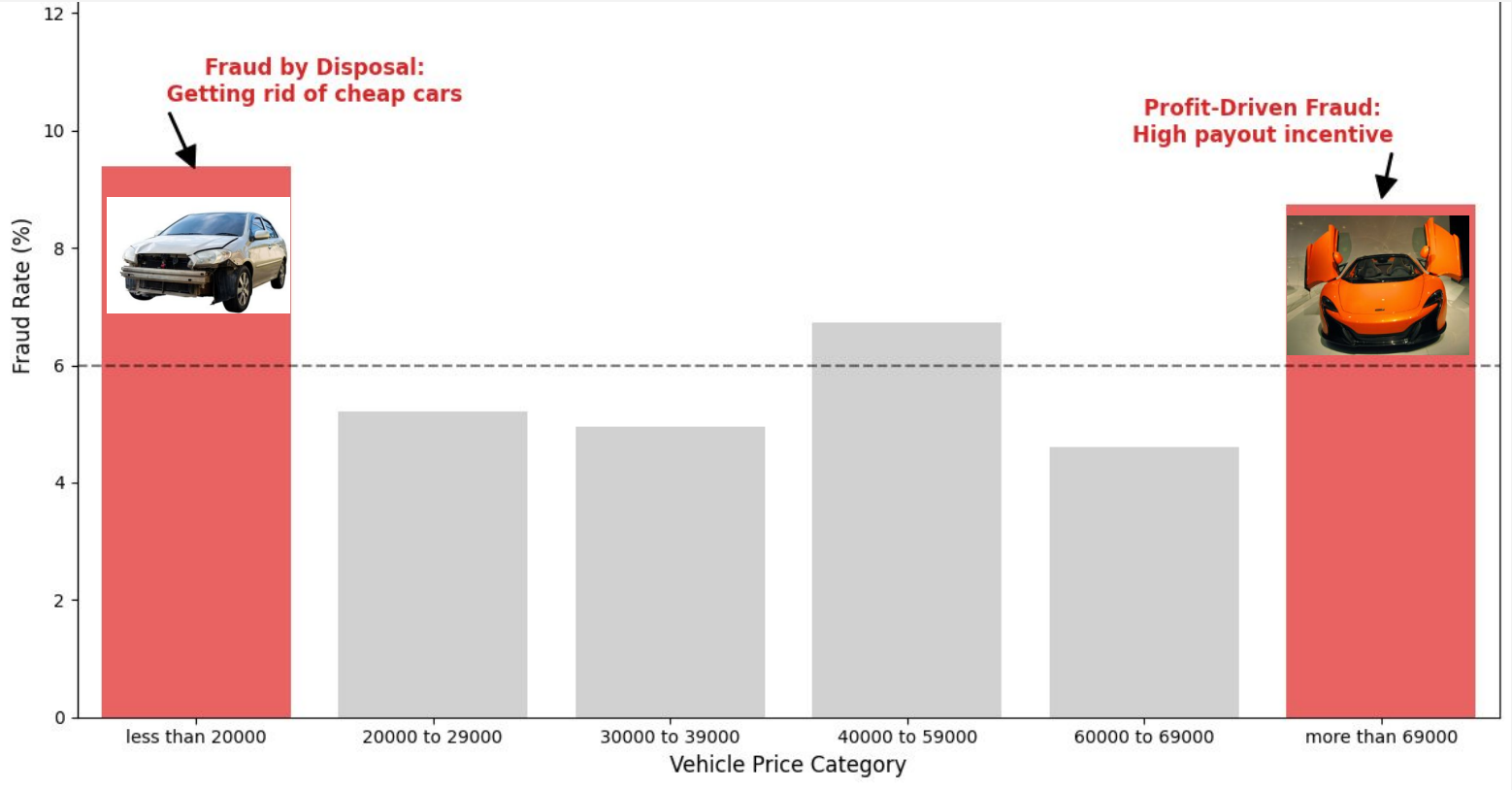
- No missing and no duplicates
- Grouping categories with <100 cases.
- Creation of new features:
  - Days to report
  - Is_Policy_Holder_Driving (dummy)
  - Weekend (dummy)
  - Luxury (dummy)

- One hot encoding for ‘VehicleCategory’
- Categorical features conversion to numerical

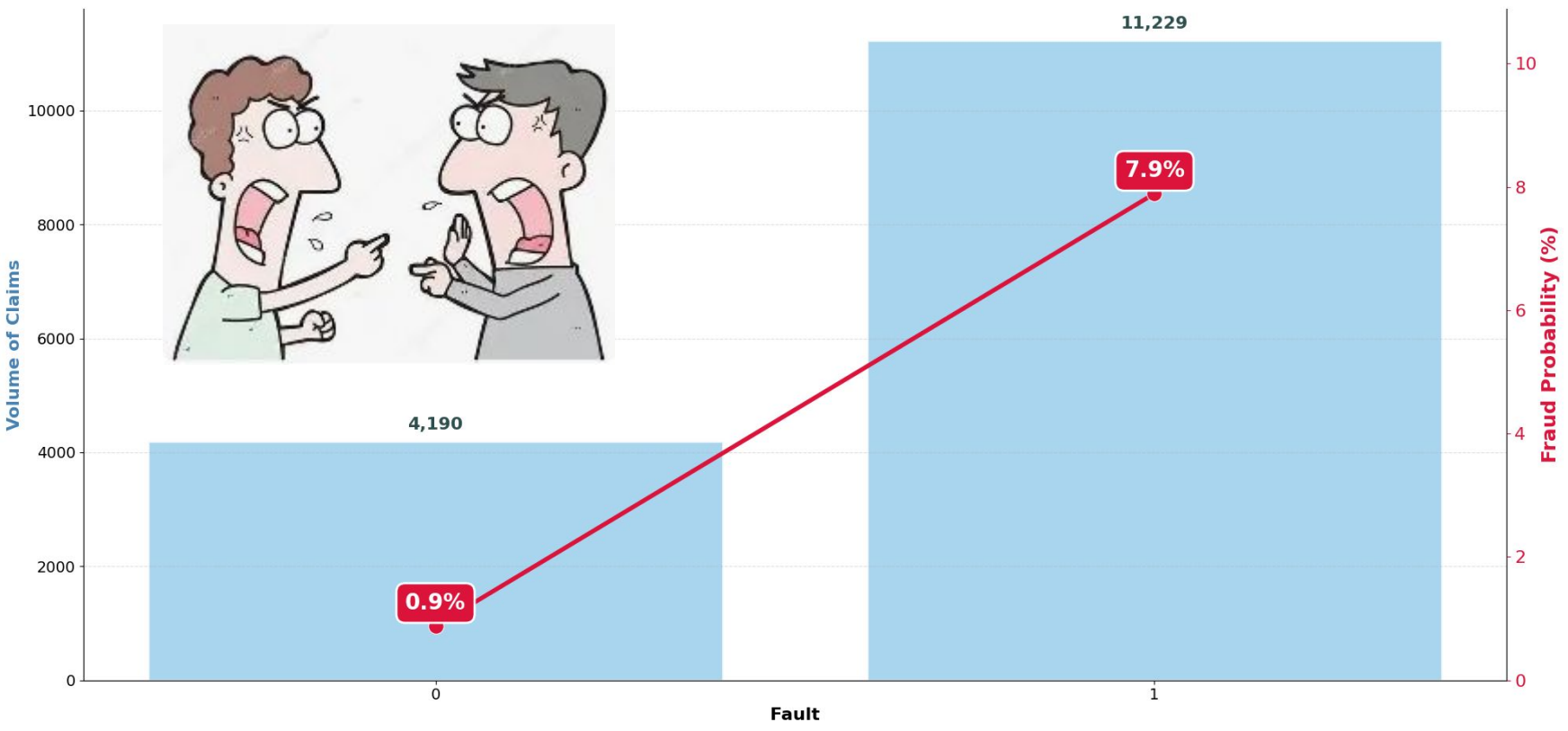
- Train and test split

- SMOTE to balance the dataset

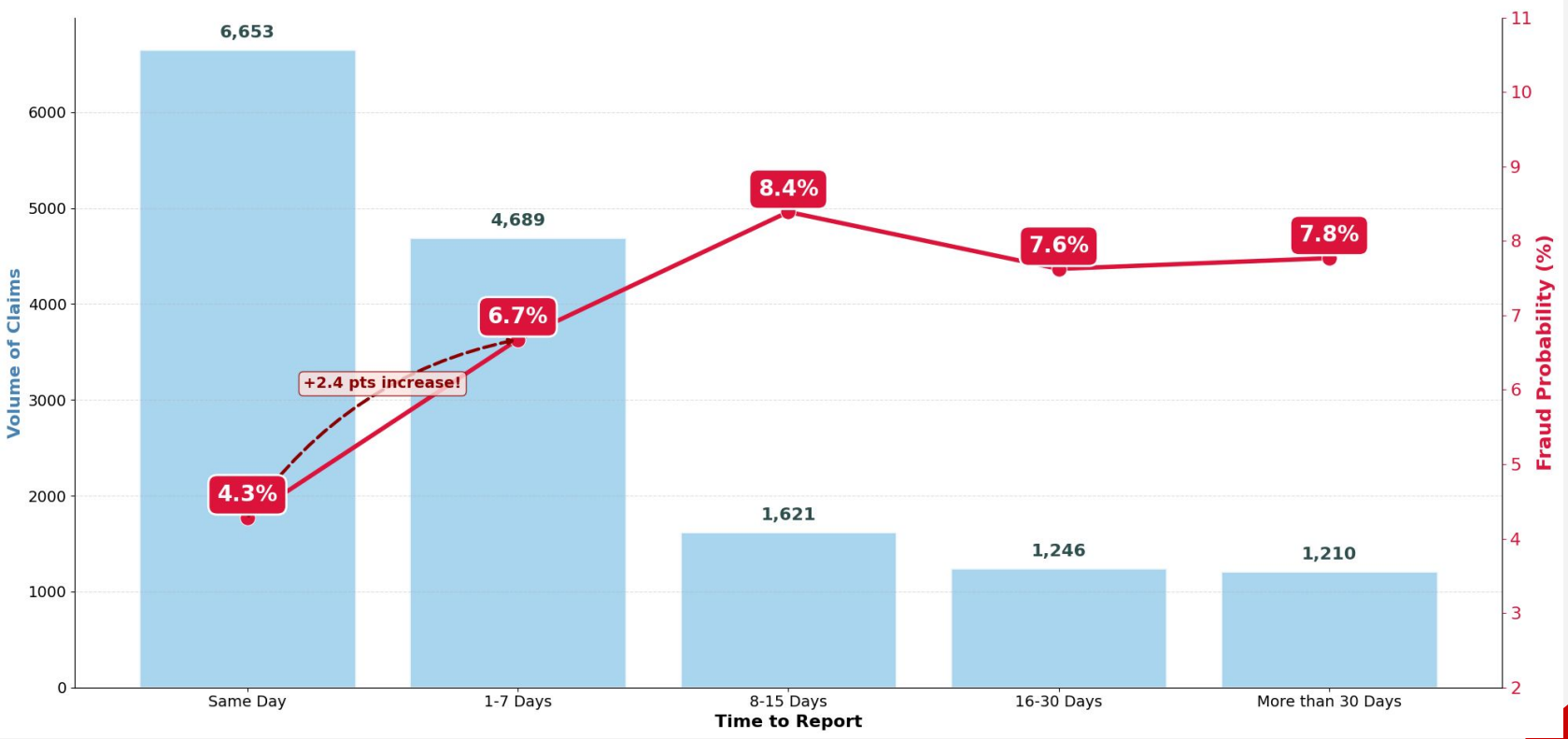
We debunked the myth that 'higher value means less crime”



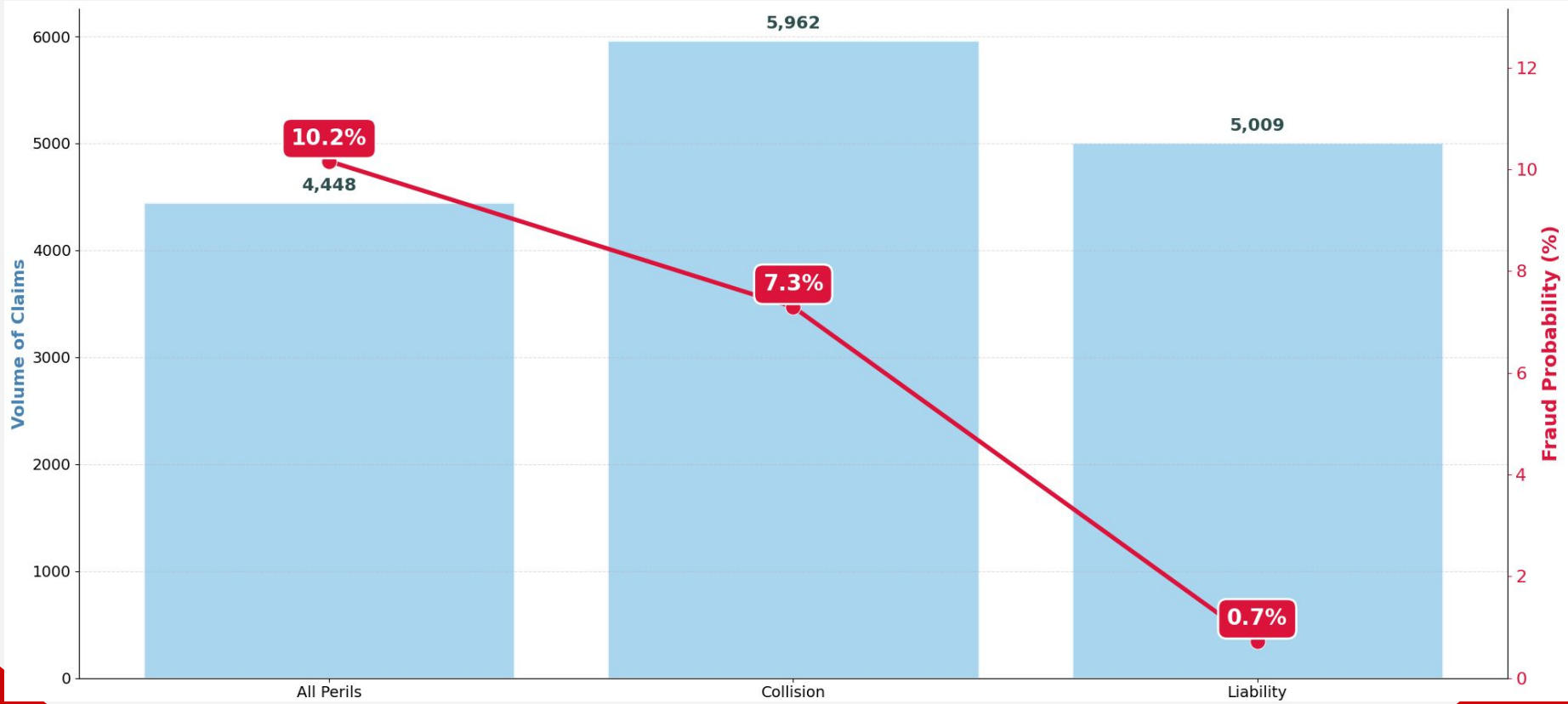
Higher Fraud Risk when Policyholder is at Fault



Time acts as a lie detector



Base Policy vs. Fraud



Models Selection & Results

	Model	AUC	Gini	Recall	Precision	F1-Score
0	Logistic Regression	0.739000	0.478000	0.648600	0.117100	0.198300
1	Random Forest	0.779400	0.558900	0.967600	0.128000	0.226200
2	XGBoost	0.802300	0.604600	0.621600	0.146300	0.236900
3	CatBoost	0.823500	0.647000	0.956800	0.127800	0.225500
4	Cat Boost Tuned	0.822900	0.645800	0.956800	0.127800	0.225500

Logistic Regression: Simple but limited. Failed to capture complex, non-linear fraud patterns.

Random Forest: Improved complexity but there are better algorithms.

XGBoost: Powerful boosting, but less efficient at handling high-cardinality categorical data.

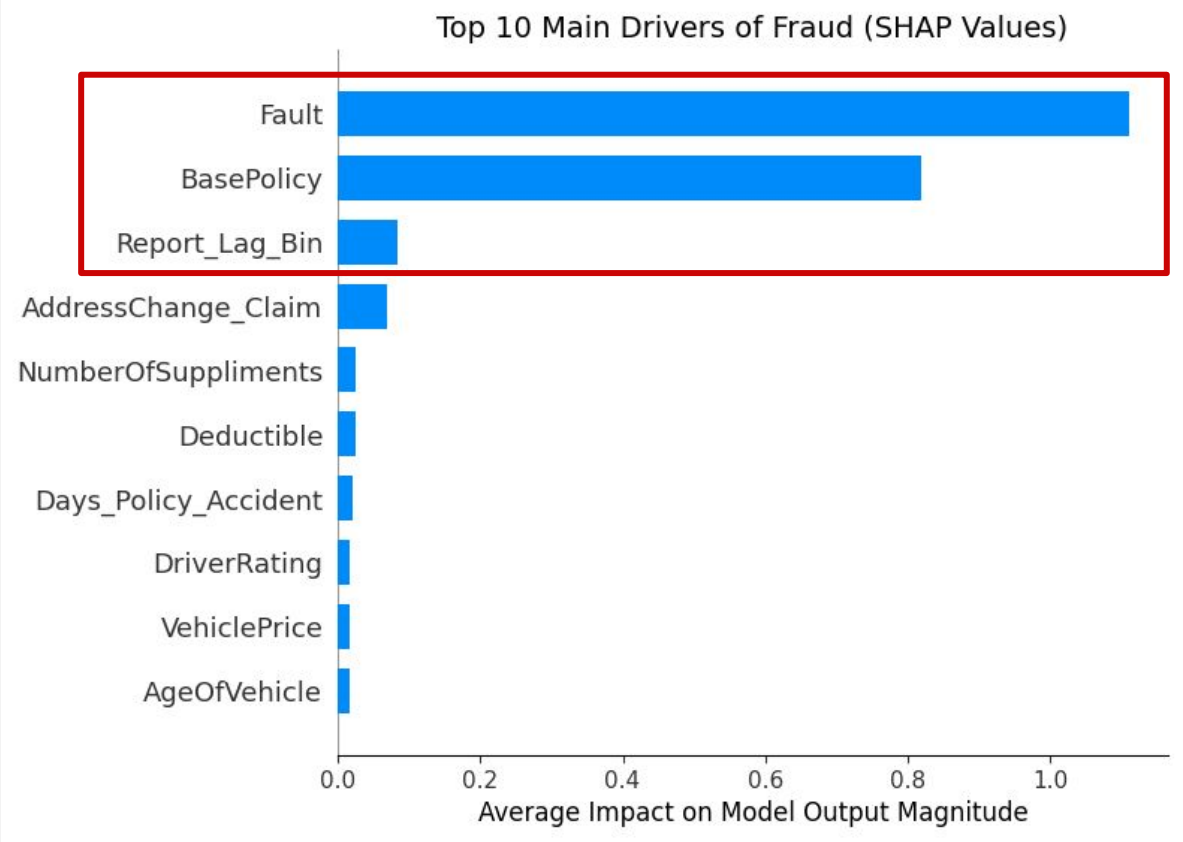
CatBoost: Best handling of categorical features and superior stability in ranking fraudulent risk.

CatBoost Tuned: Despite applying a rigorous 2-phase optimization metrics remained stable .

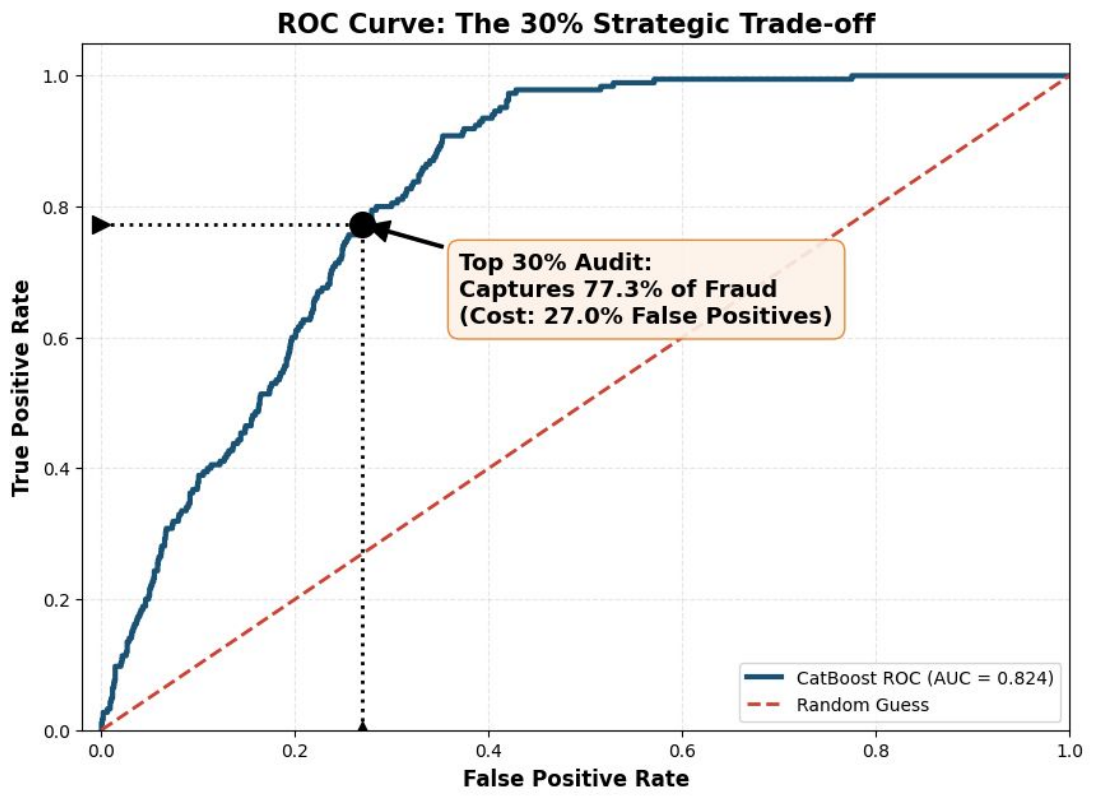
Key Hyperparameters

- **Iterations** (500): Set to ensure full model convergence while maintaining high computational efficiency.
- **Loss Function** (Logloss): Optimized for **binary classification** to provide precise probability estimates for fraud.
- **Evaluation Metric** (AUC): Used to maximize the model's ability **to rank high-risk claims** correctly.
- **Auto Class Weights** (Balanced): Crucial for handling the 6% fraud **imbalance**.

Decoding the Main Drivers of Insurance Fraud



Capturing 77% of Fraud with 30% Operational



Project Impact & Key Takeaways

1. Business Value

- Operational Efficiency
- Customer Experience (Fast-Tracking)

	Risk Tier	Score Range (Prob.)	Total Claims	Actual Fraud Cases	Fraud Rate
0	Very High	0.665 - 0.870	617	98	15.88%
1	High	0.574 - 0.665	615	70	11.38%
2	Medium	0.176 - 0.574	617	16	2.59%
3	Low	0.065 - 0.176	618	1	0.16%
4	Very Low	0.008 - 0.065	617	0	0.00%

2. Challenges & Limitations

- Unbalanced data requires special preprocessing/ hyperparameters
- This model requires continuous retraining to detect emerging fraud tactics

3. Future Work

- Test the model on more recent data
- External data intergration
- Try neural networks

Q&A